Automatic Vowel Classification in English Speech Using Formants and MFCC Features with Machine Learning

Shahd Abu Odah [1], *Yasmin Al Shawawrh* [2], Haneen Qaisi[3]

[1] Department of Electrical and Computer Engineering
[2] ENCS5344 - Spoken Language Processing
1212077@student.birzeit.edu[1], 1220848@ student.birzeit.edu[2], 1220201@student.birzeit.edu[3]

Abstract

In this project, a system was built to recognize spoken vowel sounds using computers. Two types of features were extracted from audio recordings: formants (F1 and F2), which describe how sound travels in the mouth, and MFCCs, which are commonly used in speech recognition. The effectiveness of these features was tested with different classifiers.

For formants, a simple method called Nearest Centroid was applied, and machine learning models such as Support Vector Machine (SVM) and Random Forest were also tested. For MFCCs, only machine learning models were employed. The vowel recordings were obtained from the KIDVOW dataset, which contains voices of both children and adults.

The models were trained and tested using 5-fold cross-validation to ensure reliability of the results. It was found that the best accuracy was achieved when MFCCs were combined with machine learning. In contrast, the simple formant method performed adequately but with lower accuracy. It was also observed that some vowels were recognized more easily than others.

Overall, the project demonstrates that MFCCs combined with machine learning are more effective for vowel classification and can contribute to the development of better speech recognition systems.

1. Introduction

Understanding and recognizing spoken vowels is considered an important step in building speech recognition systems. Since vowels carry a large amount of information in spoken language, their correct identification is regarded as essential for improving applications such as voice assistants, language learning apps, and speech therapy tools.

In this project, the classification of vowel sounds using machine learning was investigated. Audio recordings from the KIDVOW dataset, which includes speakers of various ages and genders, were utilized. The objective was to test and compare the performance of different features and models in recognizing vowel sounds.

Two types of features were employed:

Formants (F1 and F2): regarded as key frequency components that describe vowel shape.

Mel-Frequency Cepstral Coefficients (MFCCs): widely used in speech recognition.

Classifiers were built using these features and evaluated with 5-fold cross-validation. For formants, both a basic method (Nearest Centroid) and machine learning models (SVM and Random Forest) were tested. For MFCCs, only machine learning models were applied.

The aim was to compare the performance of these approaches and to identify which features and models achieved the best results in vowel classification.

2. Background

Speech recognition is defined as a field that focuses on enabling computers to understand human speech. A key aspect of this process is the recognition of vowel sounds, which can be performed using different types of features. One traditional approach is based on formants, particularly the first two (F1 and F2), which represent important frequency patterns in the human voice. These formants vary depending on the shape of the mouth during speech production, and they have long been employed for vowel classification.

Another widely adopted approach relies on Mel-Frequency Cepstral Coefficients (MFCCs), which are designed to mimic human auditory perception. MFCCs have been extensively applied in modern speech and audio recognition systems, as they capture detailed and informative characteristics of audio signals. To improve classification accuracy, machine learning models such as Support Vector Machine (SVM) and Random Forest have frequently been used in the analysis of these features.

In this project, both formant-based and MFCC-based vowel classification methods were explored and compared.

3. Methodology

This project focuses on the classification of spoken vowel sounds using two main types of features: formants (F1 and F2) and Mel-Frequency Cepstral Coefficients (MFCCs). The overall framework is divided into four steps: dataset collection, preprocessing, feature extraction, and classification using machine learning models.

Step 1: Dataset Collection and Preprocessing
Vowel audio samples were obtained from the UT Dallas KIDVOW dataset, which contains recordings of children and adults pronouncing isolated vowels. Each audio file (.wav) was labeled with the corresponding vowel (e.g., /i/, /a/, /u/). The file paths and labels were indexed in a CSV file (preproc_index.csv).

Before feature extraction, all signals were standardized by resampling to a fixed sampling rate and normalizing the amplitude. This ensured uniformity across the dataset.

Step 2A: Formant Feature Extraction

Formants represent resonance frequencies of the vocal tract. Using Parselmouth (a Python interface to Praat), the first two formants (F1 and F2) were extracted from each audio sample. These formants are directly related to the length and shape of the vocal tract, which varies with vowel articulation.

The resonance frequencies are computed from the Linear Predictive Coding (LPC) spectrum. The LPC model approximates the speech signal $s[n]$ as:

$$s[n] \approx -\sum_{k=1}^{p} a_k S[n-k] + Ge[n]$$

Where:

- a_k are the LPC Coefficients,
- p is the prediction order,
- G is the gain,
- e[n] is the excitation (error term)

The formants F_i are obtained from the roots of the LPC polynomial:

$$A(z) = 1 + \sum_{k=1}^{p} a_k z^{-k}$$

The angles of the complex roots give the formant frequencies in Hz:

$$F_i = \frac{\theta_i \cdot f_s}{2\pi}$$

where θ_i is the angle of the root and f_s is the sampling frequency.
The distribution of vowels in the **formant space** was visualized using a 2D scatter plot of **F2 vs. F1** (see Figure 1).

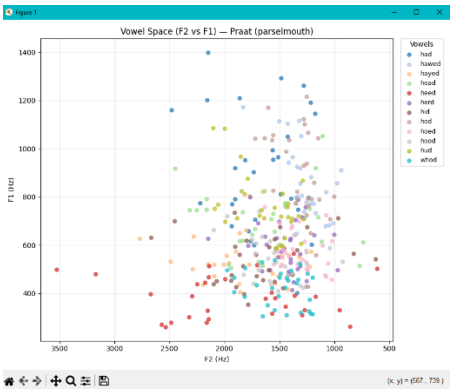

Figure 1: Vowel space (F2 vs F1)

Step 2B: MFCC Feature Extraction

Mel-Frequency Cepstral Coefficients (MFCCs) were extracted to capture perceptually relevant information from speech. Each audio signal was divided into short overlapping frames, windowed with a Hamming function, and transformed into the frequency domain using the **Discrete Fourier Transform (DFT).**
The power spectrum was then passed through a **Mel filterbank**, which maps frequency f (Hz) to the Mel scale:

$$M(f) = 2595 \cdot log_{10}\left(1 + \frac{f}{700}\right)$$

The log-energy outputs from the filterbank were decorrelated using the **Discrete Cosine Transform (DCT).**

These coefficients form a compact representation of the speech spectrum, with the first 12–13 coefficients typically used in classification. An example MFCC spectrogram is shown in **Figure 2**, illustrating the time-frequency representation of vowel sounds.

Step 3: Classification

The extracted features (formants and MFCCs) were used as input to classification models. For formants, both a **basic classifier (Nearest Centroid)** and machine learning models (**Support Vector Machine – SVM, Random Forest**) were applied. For MFCCs, only machine learning models were used.
To ensure robustness, **5-fold cross-validation** was performed, where the dataset was split into 5 subsets, and the models were trained and tested iteratively. The average accuracy across folds was used as the final performance measure

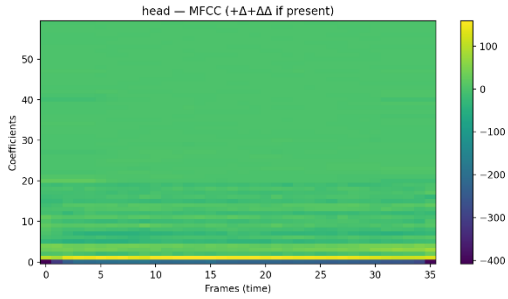

Figure 2: MFCCs of "head" with 39 coefficients.

4. Experiments and Results

In this project, experiments were conducted to evaluate the effectiveness of vowel classification using two types of features: formant frequencies (F1 and F2) and Mel-Frequency Cepstral Coefficients (MFCCs). The goal was to determine which feature set and classification method yielded the highest accuracy across five vowel classes.

Dataset

A subset of the UTDallas KIDVOW vowel dataset was used, containing .wav recordings of spoken vowels by children. Each audio file was labeled with one of the five English vowels: /a/, /e/, /i/, /o/, or /u/. The audio files were preprocessed, and relevant features were extracted as outlined in the methodology section.

Evaluation Metrics

To ensure a robust and unbiased evaluation, 5-fold stratified cross-validation was applied. Accuracy was used as the primary performance metric. Additionally, confusion matrices were generated to visualize the distribution of predictions and identify common misclassifications.
Accuracy was calculated using the following formula:

$$\text{Accuracy} = \frac{Number\ Of\ Correct\ Predictions}{Total\ Number\ Of\ Samples}$$

Confusion Matrix was used to analyze classification errors across different vowel classes.

Experiments Performed

Several experiments were carried out using different combinations of features and classifiers, as summarized in the table below:

Feature Set	Classifier	Purpose
Formants (F1, F2)	Nearest Centroid	Baseline (no ML)
Formants (F1, F2)	SVM	Supervised ML comparison
Formants (F1, F2)	Random Forest	Supervised ML comparison
MFCC	SVM	Feature-rich + ML
MFCC	Random Forest	Feature-rich + ML

Table 1: Summery Of Experiments

Quantitative Results

The classification accuracies for each combination of features and classifiers are shown in the figure below.

```
Formants (No ML) Accuracy: 0.319
Formants SVM Accuracy: 0.344
Formants RF Accuracy: 0.272
MFCC SVM Accuracy: 0.506
MFCC RF Accuracy: 0.500
```

Figure 3: Accuracy results for different models.

As seen in the results, the MFCC-based classifiers outperformed the formant-based ones. The highest accuracy (50.6%) was achieved by the MFCC + SVM model, followed closely by MFCC + Random Forest (50.0%). The lowest accuracy was recorded when using formants with Random Forest, while the Nearest Centroid classifier (no ML) achieved a baseline accuracy of 31.9%.

MFCC Heatmap Visualization

To provide further insight into the extracted MFCC features, heatmaps were generated for individual vowel samples. These heatmaps represent the MFCC coefficient values over time, offering a visual pattern of how each vowel's spectral properties evolve. An example is shown below for the vowel "whod", where the x-axis indicates the time frames, the y-axis represents MFCC coefficients, and the color intensity denotes the coefficient magnitude.

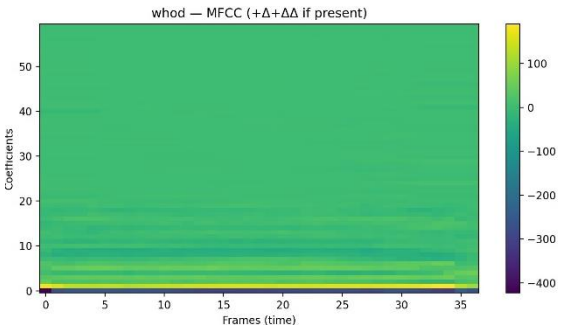

Figure 4: MFCCs of "whod" with 39 coefficients

Such visualizations help highlight differences in spectral structure between vowels, aiding both feature understanding and classifier performance analysis.

Qualitative Results

To further analyze classifier behavior, confusion matrices were visualized. The figure below shows the confusion matrix for the best-performing configuration: MFCC with SVM.

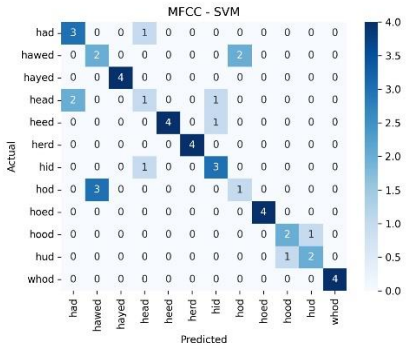

Figure 5: Confusion matrix for MFCC with SVM.

It was observed that MFCC-based models helped improve classification between similar vowels, such as /e/ and /i/, which were more often confused when using only formants.

5. Conclusion and Future Work

In this project, vowel classification was investigated using two main types of features: formants (F1 and F2) and Mel-Frequency Cepstral Coefficients (MFCCs). Various classification methods were applied, including Nearest Centroid (without machine learning) and machine learning-based models such as Support Vector Machines (SVM) and Random Forests (RF). The experiments were conducted using 5-fold cross-validation to ensure robustness.

From the results, it was concluded that MFCC features significantly outperformed formant features across all classification methods. The MFCC + SVM combination achieved the best accuracy of 50.6%, while formant-based classifiers performed less effectively, with the highest formant-based accuracy reaching only 34.4%. This demonstrates the superiority of MFCCs in capturing the spectral characteristics needed for vowel discrimination.

As for future work, several improvements can be explored. Data augmentation techniques could be applied to increase the dataset size and diversity, which may improve classifier generalization. Additionally, more advanced deep learning models such as convolutional neural networks (CNNs) or recurrent neural networks (RNNs) could be implemented to learn more complex temporal patterns in speech. Finally, extracting more MFCC derivatives (Δ and ΔΔ) or experimenting with other audio features (e.g., PLP or spectrograms) might further enhance performance.

6. References

[1] UTDallas KIDVOW Vowel Dataset. [Online]. Available: https://personal.utdallas.edu/~assmann/KIDVOW/

[2] Shaaban, A. Vowel classification using formants and MFCCs. [Online]. Available: https://www.kaggle.com/code/abdelrahmansha/vowel-disting

[3] PyCharm IDE. JetBrains. [Tool used].

[4] Visual Studio Code. Microsoft. [Tool used].

[5] Python Programming Language. [Tool used].

[6] Instructor Lecture Slides, Speech Processing Course, Birzeit University, 2025.